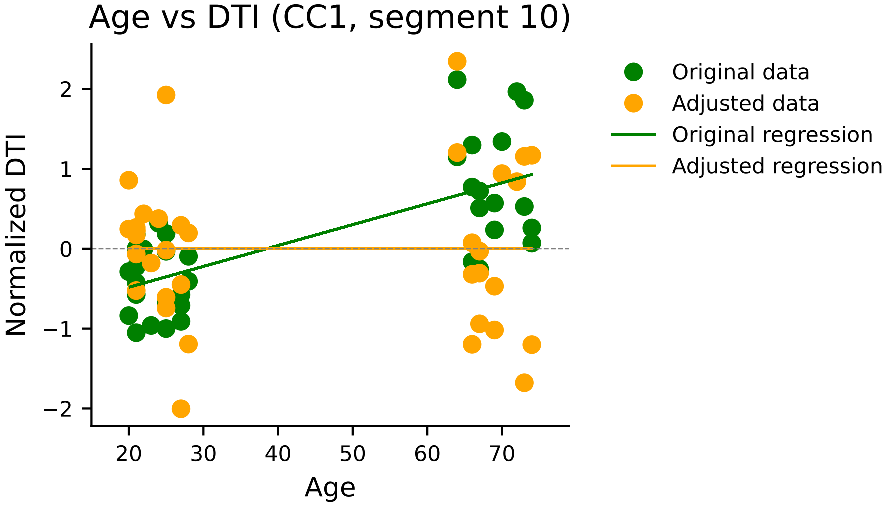
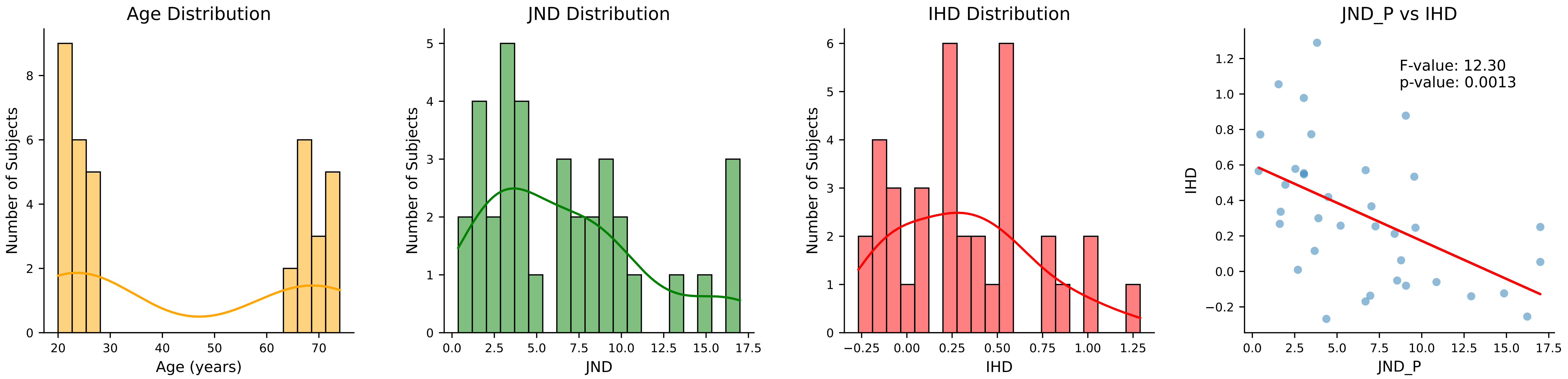
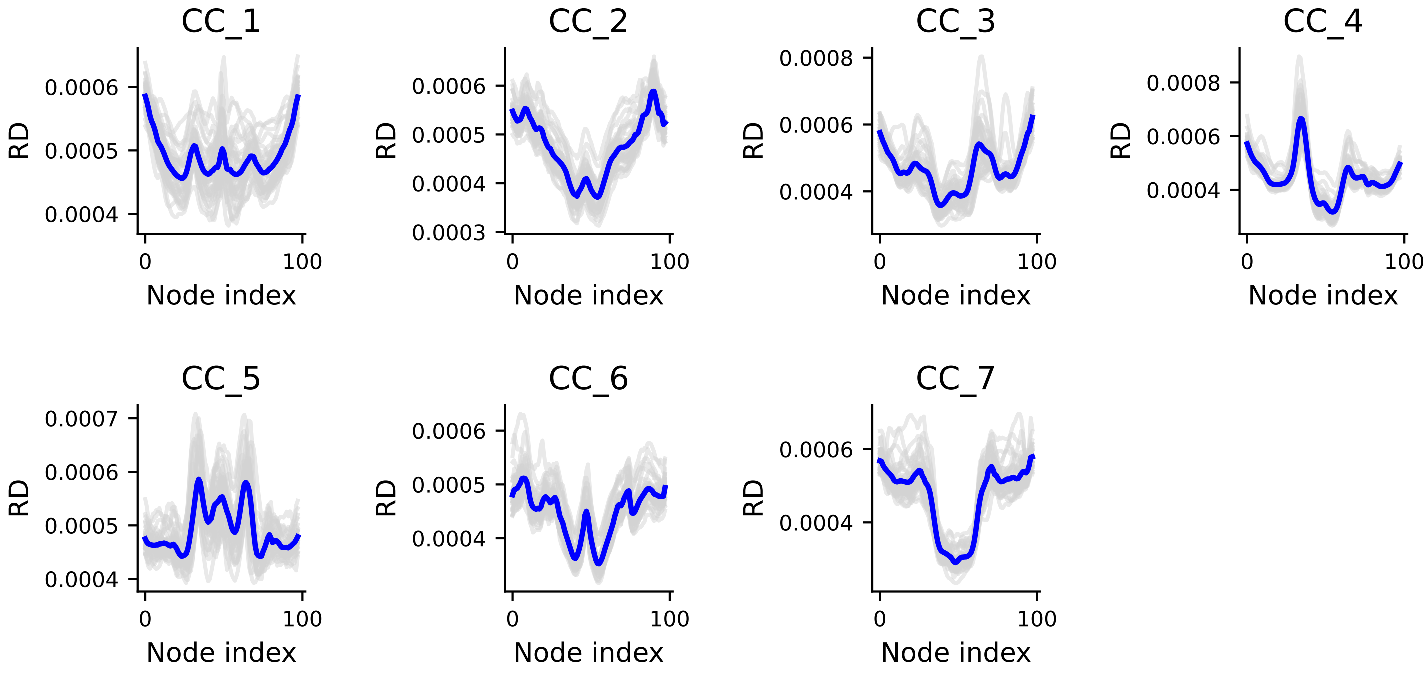
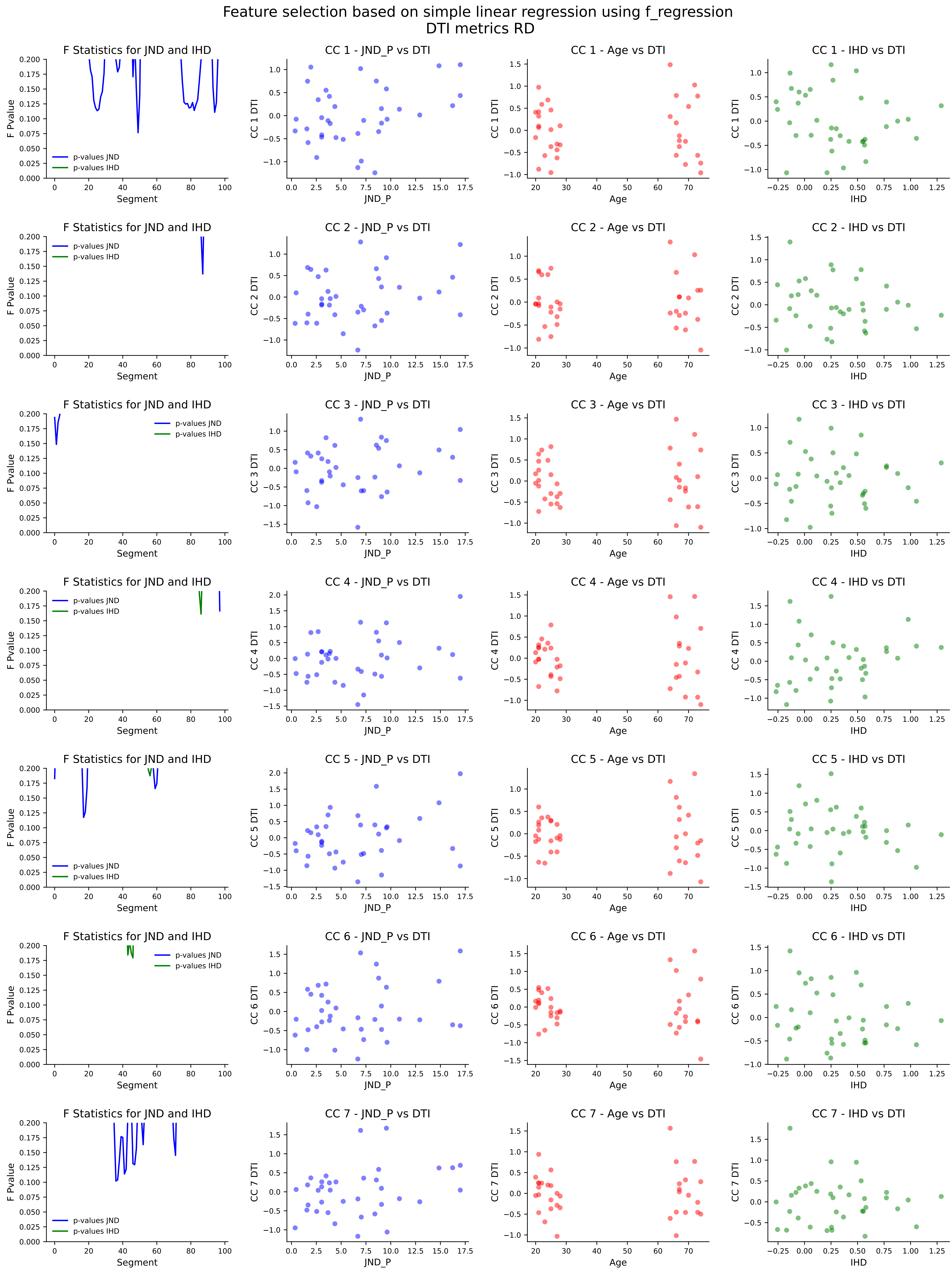


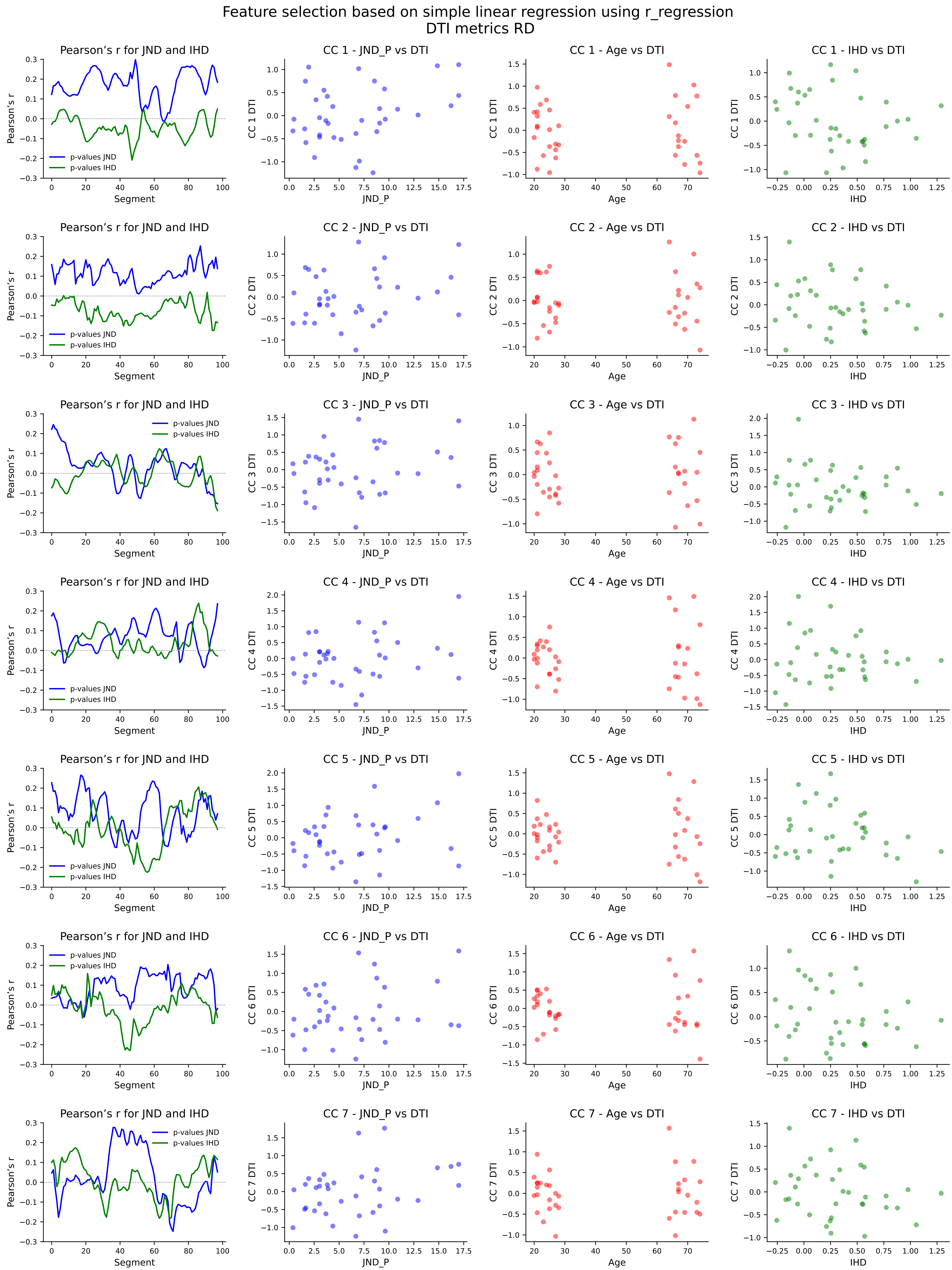
Original : p-value = 0.000, F-stat = 34.90

Adjusted : p-value = 1.000, F-stat = 0.00

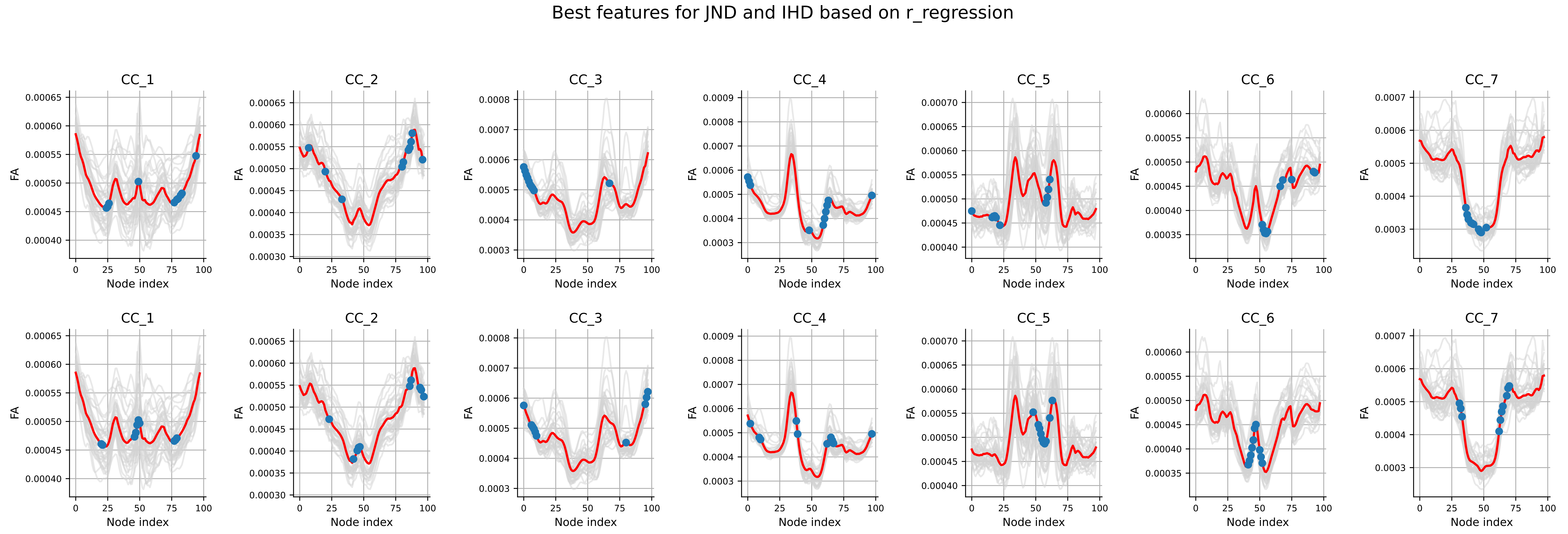

Age Distribution
Number of Subjects
Age (years)
JND Distribution
Number of Subjects
JND
IHD Distribution
Number of Subjects
IHD
JND_P vs IHD
F-value: 12.30
p-value: 0.0013
IHD
JND_P

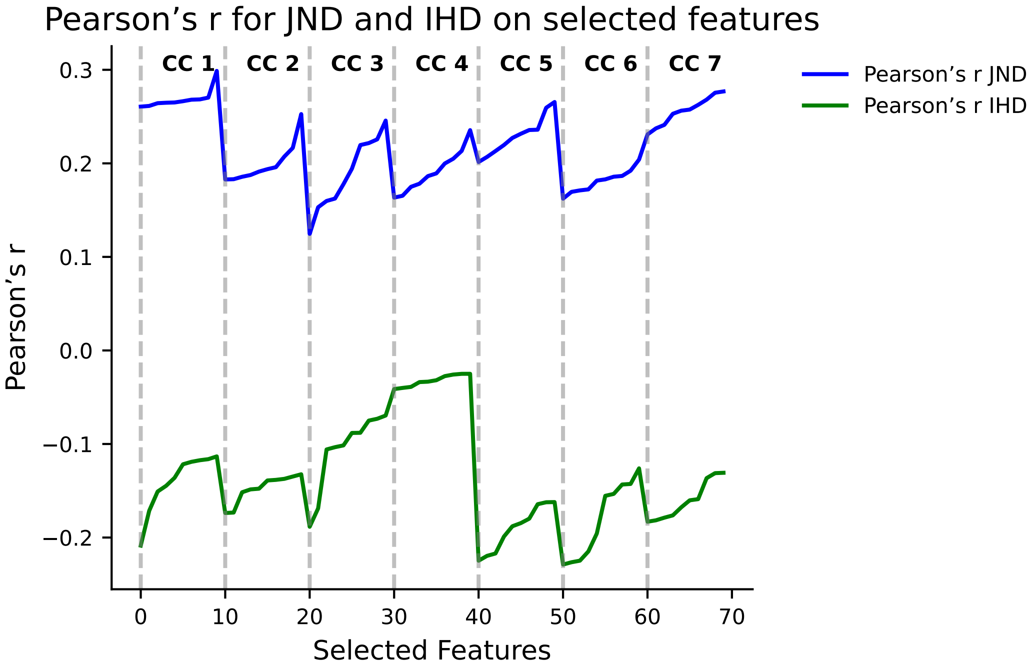

CC_1
CC_2
CC_3
CC_4
CC_5
CC_6
CC_7
RD
Node index
0
100
0.0003
0.0004
0.0005
0.0006
0.0007
0.0008


Feature selection based on simple linear regression using f_regression
DTI metrics RD
F Statistics for JND and IHD
CC 1 - JND_P vs DTI
CC 1 - Age vs DTI
CC 1 - IHD vs DTI
F Statistics for JND and IHD
CC 2 - JND_P vs DTI
CC 2 - Age vs DTI
CC 2 - IHD vs DTI
F Statistics for JND and IHD
CC 3 - JND_P vs DTI
CC 3 - Age vs DTI
CC 3 - IHD vs DTI
F Statistics for JND and IHD
CC 4 - JND_P vs DTI
CC 4 - Age vs DTI
CC 4 - IHD vs DTI
F Statistics for JND and IHD
CC 5 - JND_P vs DTI
CC 5 - Age vs DTI
CC 5 - IHD vs DTI
F Statistics for JND and IHD
CC 6 - JND_P vs DTI
CC 6 - Age vs DTI
CC 6 - IHD vs DTI
F Statistics for JND and IHD
CC 7 - JND_P vs DTI
CC 7 - Age vs DTI
CC 7 - IHD vs DTI
p-values JND
p-values IHD
F Pvalue
Segment
JND_P
Age
IHD
CC 1 DTI
CC 2 DTI
CC 3 DTI
CC 4 DTI
CC 5 DTI
CC 6 DTI
CC 7 DTI

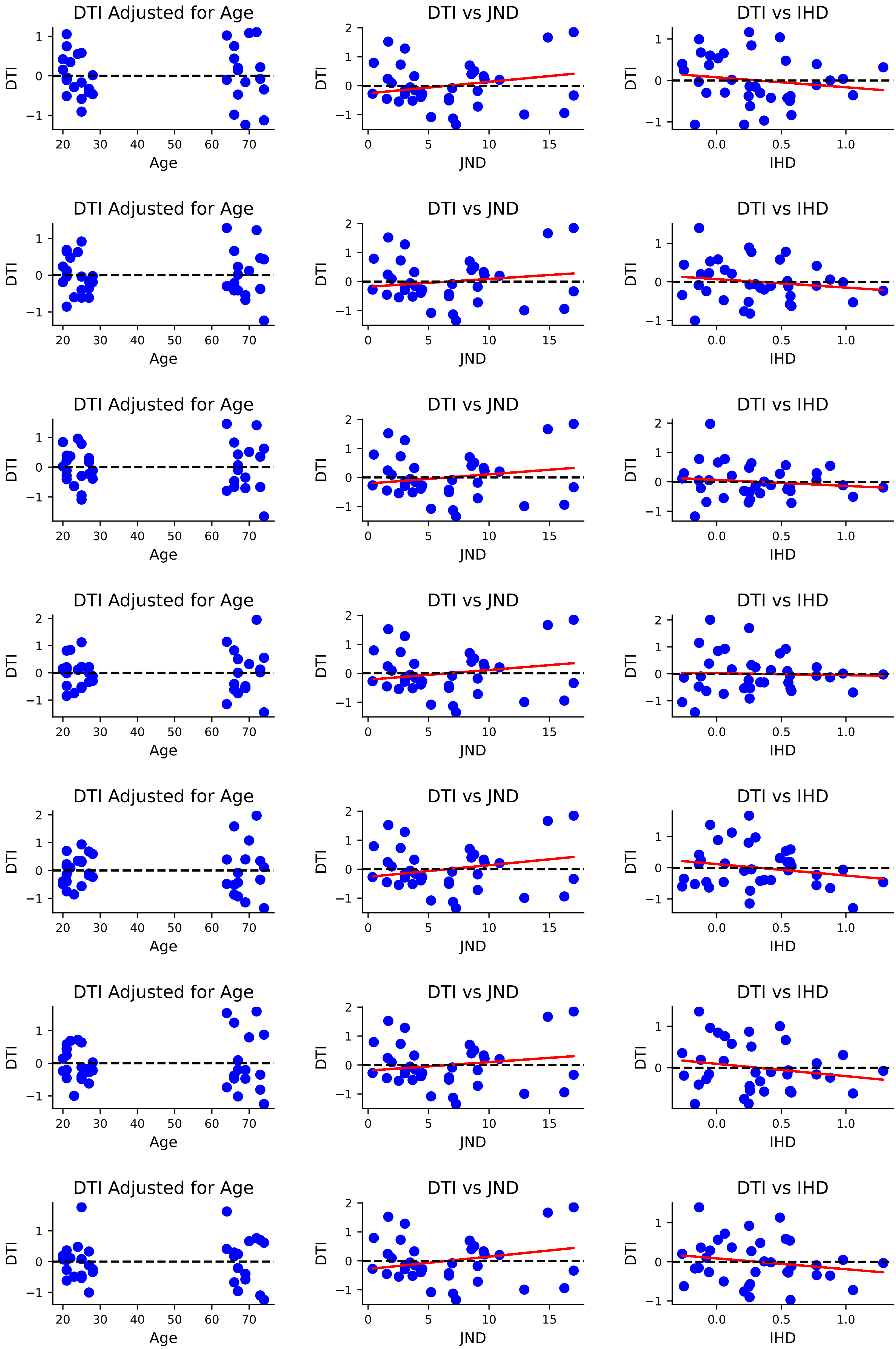


Feature selection based on simple linear regression using r_regression
DTI metrics RD
Pearson's r for JND and IHD
CC 1 - JND_P vs DTI
CC 1 - Age vs DTI
CC 1 - IHD vs DTI
Pearson's r for JND and IHD
CC 2 - JND_P vs DTI
CC 2 - Age vs DTI
CC 2 - IHD vs DTI
Pearson's r for JND and IHD
CC 3 - JND_P vs DTI
CC 3 - Age vs DTI
CC 3 - IHD vs DTI
Pearson's r for JND and IHD
CC 4 - JND_P vs DTI
CC 4 - Age vs DTI
CC 4 - IHD vs DTI
Pearson's r for JND and IHD
CC 5 - JND_P vs DTI
CC 5 - Age vs DTI
CC 5 - IHD vs DTI
Pearson's r for JND and IHD
CC 6 - JND_P vs DTI
CC 6 - Age vs DTI
CC 6 - IHD vs DTI
Pearson's r for JND and IHD
CC 7 - JND_P vs DTI
CC 7 - Age vs DTI
CC 7 - IHD vs DTI
Pearson's r
Segment
p-values JND
p-values IHD
CC 1 DTI
CC 2 DTI
CC 3 DTI
CC 4 DTI
CC 5 DTI
CC 6 DTI
CC 7 DTI
JND_P
Age
IHD


Best features for JND and IHD based on r_regression
CC_1
CC_2
CC_3
CC_4
CC_5
CC_6
CC_7
FA
Node index


Pearson’s r for JND and IHD on selected features
CC 1
CC 2
CC 3
CC 4
CC 5
CC 6
CC 7
Pearson’s r JND
Pearson’s r IHD
Pearson’s r
0.3
0.2
0.1
0.0
−0.1
−0.2
0
10
20
30
40
50
60
70
Selected Features

Linear regression of DTI on JND and IHD after adjustment for age
Mean of 10 best features DTI metrics RD



JND RD

RD JND \ segment cc: 1 [49, 94, 25, 82, 26, 79, 24, 80, 83, 77]
p-value pour cc 1: 0.0183
p-value bonf. corrected pour cc 1: 0.128 □

RD JND \ segment cc: 2 [87, 86, 81, 96, 85, 80, 33, 7, 88, 20]
p-value pour cc 2: 0.0926
p-value bonf. corrected pour cc 2: 0.648 □

RD JND \ segment cc: 3 [1, 2, 0, 3, 4, 5, 6, 7, 97, 8]
p-value pour cc 3: 0.1
p-value bonf. corrected pour cc 3: 0.701 □

RD JND \ segment cc: 4 [97, 61, 60, 62, 1, 59, 63, 0, 48, 2]
p-value pour cc 4: 0.0813
p-value bonf. corrected pour cc 4: 0.569 □

RD JND \ segment cc: 5 [17, 18, 59, 19, 60, 0, 58, 16, 61, 22]
p-value pour cc 5: 0.0447
p-value bonf. corrected pour cc 5: 0.313 □

RD JND \ segment cc: 6 [68, 52, 55, 53, 54, 93, 92, 66, 56, 75]
p-value pour cc 6: 0.132
p-value bonf. corrected pour cc 6: 0.923 □

RD JND \ segment cc: 7 [36, 37, 41, 42, 47, 46, 38, 71, 48, 52]
p-value pour cc 7: 0.0199
p-value bonf. corrected pour cc 7: 0.139 □

____________________IHD RD____________________

RD \ segment cc: 1 [47, 48, 46, 49, 78, 79, 77, 50, 21, 20]
p-value pour cc 1: 0.229
p-value bonf. corrected pour cc 1: 1.602 □

RD \ segment cc: 2 [94, 95, 42, 46, 45, 23, 87, 47, 97, 86]
p-value pour cc 2: 0.219
p-value bonf. corrected pour cc 2: 1.535 □

RD \ segment cc: 3 [97, 96, 63, 64, 62, 95, 9, 8, 65, 38]
p-value pour cc 3: 0.896
p-value bonf. corrected pour cc 3: 6.274 □

RD \ segment cc: 4 [86, 85, 88, 87, 84, 83, 28, 27, 29, 89]
p-value pour cc 4: 0.116
p-value bonf. corrected pour cc 4: 0.813 □

RD \ segment cc: 5 [56, 55, 57, 86, 54, 85, 58, 48, 84, 53]
p-value pour cc 5: 0.435
p-value bonf. corrected pour cc 5: 3.046 □

RD \ segment cc: 6 [46, 43, 45, 44, 42, 21, 51, 47, 52, 41]
p-value pour cc 6: 0.198
p-value bonf. corrected pour cc 6: 1.387 □

RD \ segment cc: 7 [69, 63, 70, 32, 14, 13, 64, 15, 62, 31]
p-value pour cc 7: 0.644
p-value bonf. corrected pour cc 7: 4.509 □